



Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW :- 2

Date: 20-04-2026

Movie Booking System

SDG 9 – Industry, Innovation and Infrastructure

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- Traditional movie ticket booking involves long queues, manual record-keeping, and frequent booking errors at counters.

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- Manual seat allocation leads to double-bookings, overbooking, and poor customer experience.

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- This project builds a relational database-driven web system to manage movies, shows, seats, users, and bookings.

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- It updates seat availability in real-time during bookings and cancellations, preventing conflicts.

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- Overall, it provides a scalable, reliable platform that improves convenience, reduces errors, and enhances the movie-going experience.

PROBLEM STATEMENT

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➤ Most cinema halls still use manual or paper-based booking systems with no real-time seat tracking.

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➤ This causes double-bookings, unavailable seat information, and long queues at ticket counters.

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➤ Errors in recording bookings and cancellations result in seat mismatches and revenue loss for theatre owners.

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➤ Lack of a centralized system makes it difficult to manage multiple screens, shows, and showtimes efficiently.

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➤ Therefore, an automated and reliable movie booking system is needed to improve accuracy and customer satisfaction.

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OBJECTIVE(S)

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➤ To design and develop a web-based movie booking system that enables users to browse movies, view available showtimes, and book seats online in real-time.

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➤ To implement automated seat availability updates and booking confirmations to eliminate double-bookings and improve the overall user booking experience.

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PROPOSED SYSTEM

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- A relational database-driven web application that manages movies, theatres, screens, showtimes, seats, users, and bookings efficiently.

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- It provides real-time seat availability tracking, ensuring accurate booking status for every show at all times.

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- The system prevents double-bookings by automating seat locking and maintaining data consistency using primary and foreign keys.

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- It solves issues like overbooking and seat conflicts by giving instant booking confirmation and availability alerts.

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- It improves operational efficiency by reducing the time required to browse, select, and book movie tickets online.

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- The system enables better business insights through reports on booking trends, popular movies, and revenue data.

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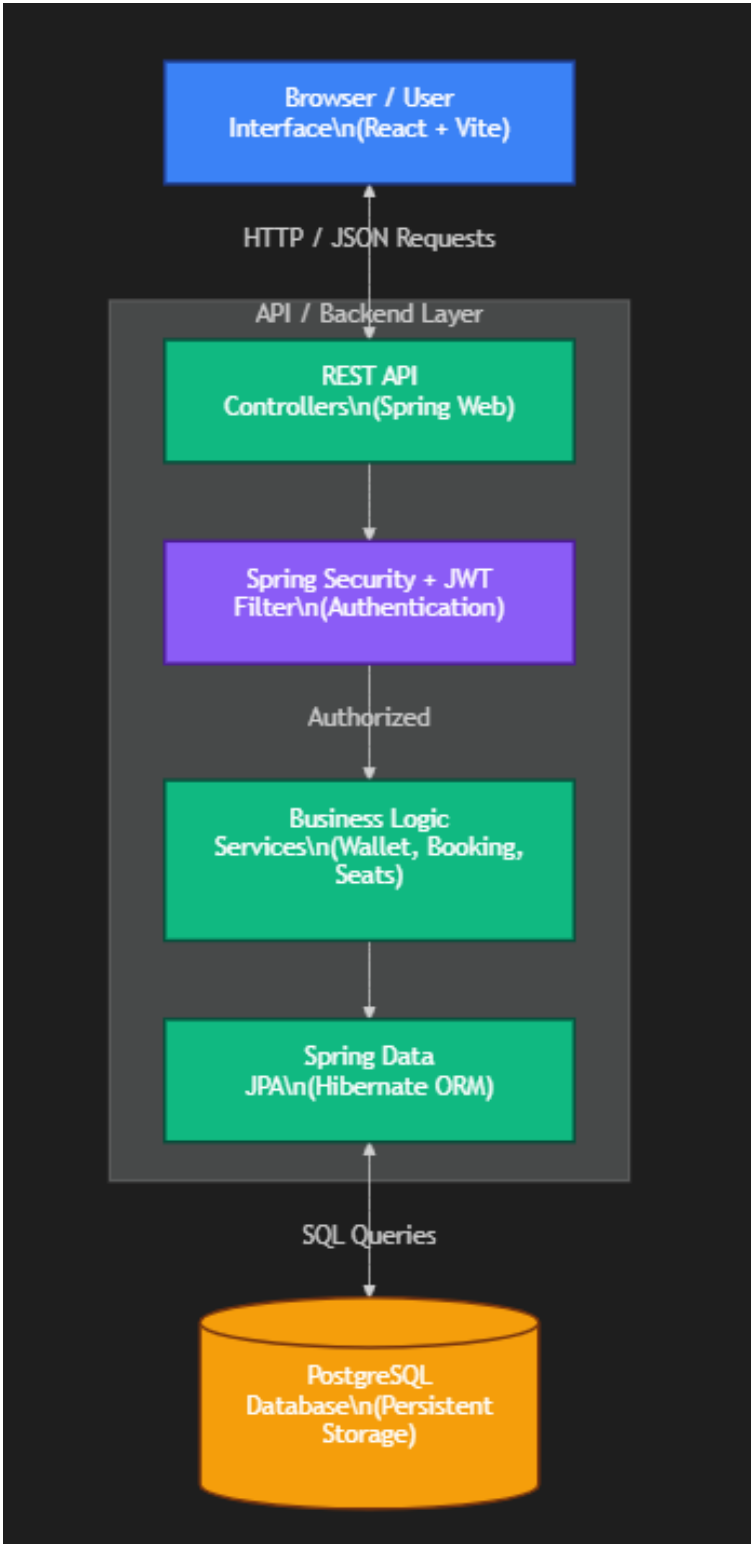
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- It offers scalability, allowing the platform to handle a large number of concurrent users and bookings without performance issues.
- Compared to traditional methods, it is more reliable, secure, and easily integrable with payment gateways and notification services.

PROPOSED SYSTEM ARCHITECTURE

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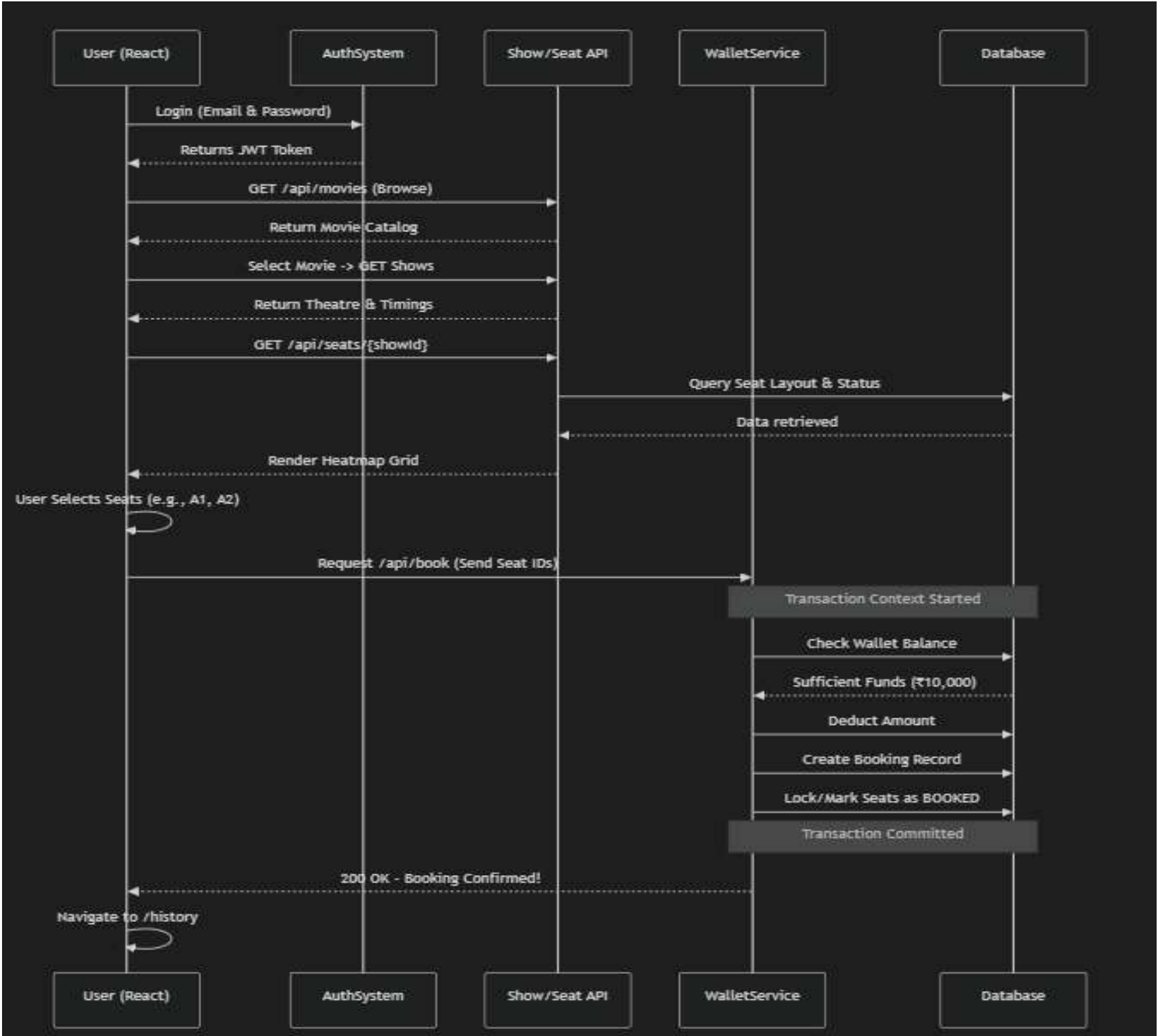
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WORKFLOW

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TECHNOLOGIES USED

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Frontend:

HTML, CSS, JavaScript, ReactJs

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Backend:

Spring boot, SpringDataJPA,Hibernate

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Database:

PostgreSQL

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Tools:

VS Code, Postman, pgAdmin, Git, GitHub, DBeaver

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Hardware Requirements:

Laptop/Desktop, 4GB RAM (8GB recommended), Internet

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Software Requirements:

Node.js, PostgreSQL, Web Browser (Chrome), VS Code

MODULE DESCRIPTION



Module 1: User Authentication

- 1 Handles user registration, login, and secure session management using JWT tokens.
- 2 Passwords are encrypted using bcrypt for security.
- 3 Role-based access differentiates admin and regular user capabilities.

Module 2: Movie & Show Management

- 4 Admins can add/edit/delete movies, theatres, screens, and showtimes.
- 5 Supports multiple screen management within a single theatre.
- 6 Ensures show schedules do not overlap on the same screen.

Module 3: Seat Booking & Availability

- 7 Displays available, booked, and locked seats for each showtime.
- 8 Prevents double-booking using database-level constraints.
- 9 Updates seat status in real-time after each booking or cancellation.

Module 4: Database Management

- Stores users, movies, theatres, shows, seats, and booking records.
- Ensures data consistency and integrity.
- Supports efficient data retrieval and updates..

IMPLEMENTATION SCREENSHOTS

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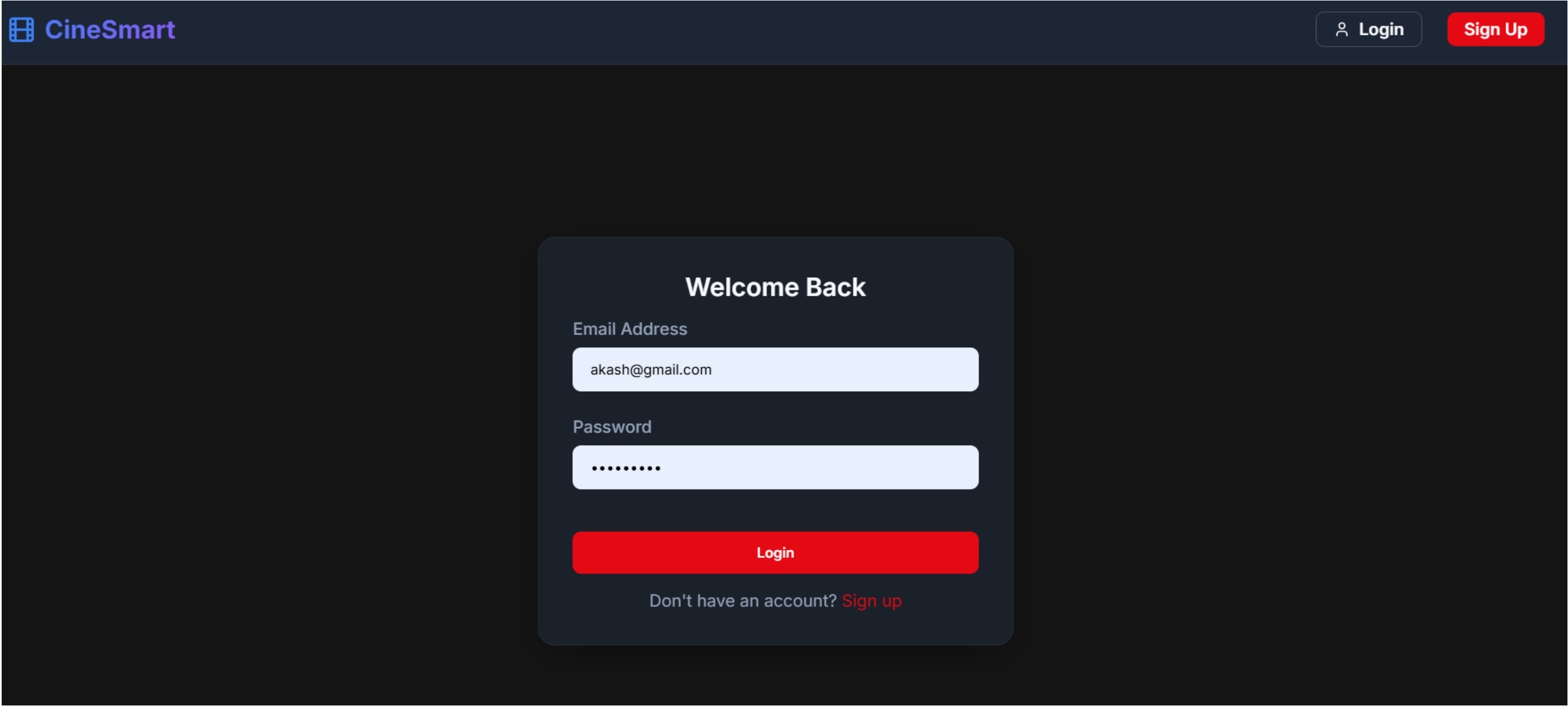
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IMPLEMENTATION SCREENSHOTS

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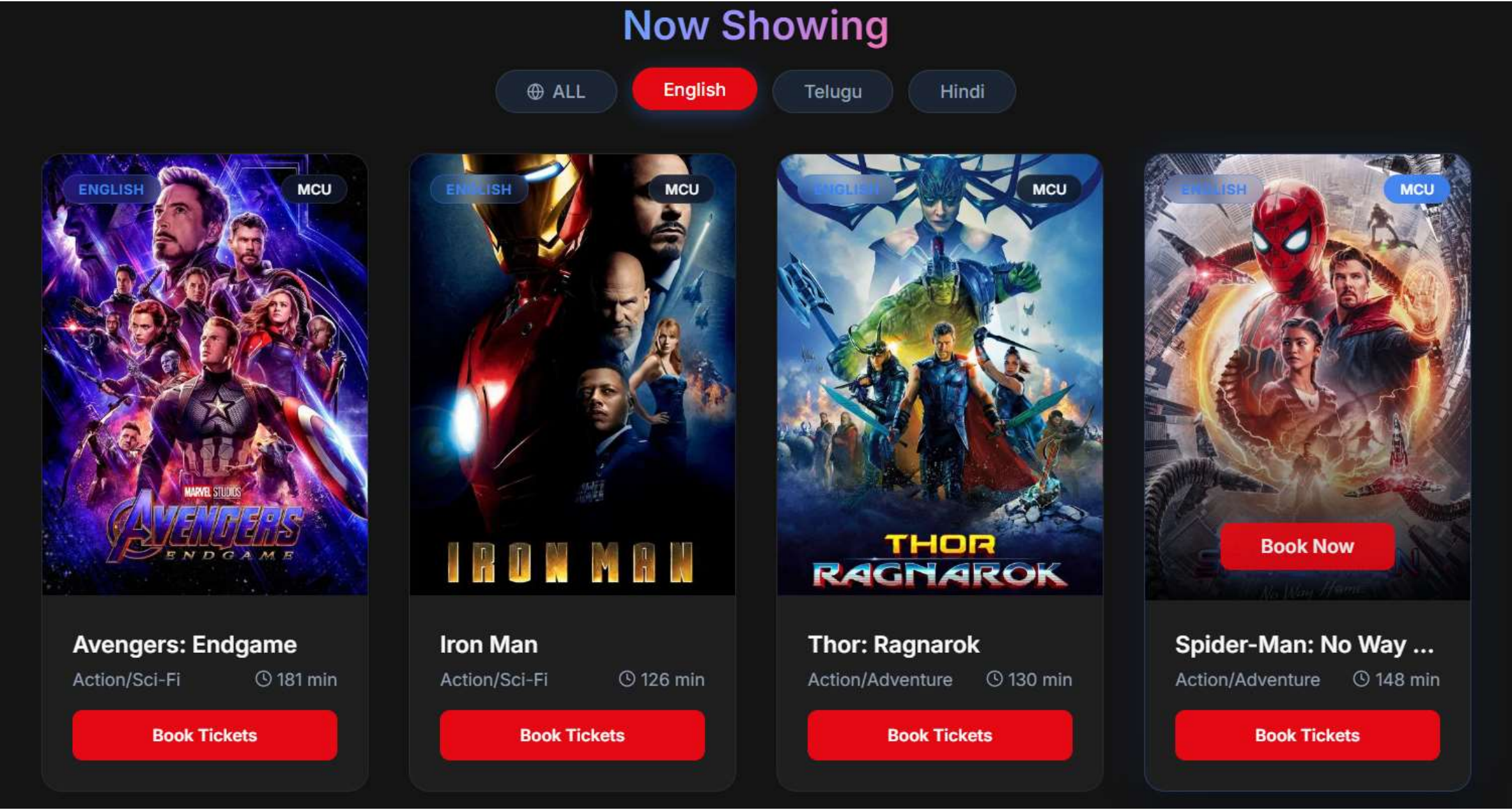
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IMPLEMENTATION SCREENSHOTS

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Baahubali 2

Baahubali 2

Tue, 21 Apr

02:30 pm

Inox Leisure Hyderabad

C1

C2

C3

C4

C5

C6

C7

C8

CONFIRMED

Booking #8

20 Apr 2026 at 08:07 am

Total Amount Paid

₹1200.00

Standard WALLET

Cancel Ticket

Dangal

Dangal

Tue, 21 Apr

02:30 pm

PVR Cinemas Hyderabad

E4

CONFIRMED

Booking #7

20 Apr 2026 at 07:52 am

Total Amount Paid

₹150.00

Standard WALLET

Cancel Ticket

Salaar: Part 1

Salaar: Part 1

Tue, 21 Apr

02:30 pm

PVR Cinemas Vizag

A2

B4

C4

C5

C6

D4

E4

E5

E6

F5

F6

G2

G4

G5

G6

H6

CONFIRMED

Booking #6

20 Apr 2026 at 07:11 am

Total Amount Paid

₹2400.00

Standard WALLET

Cancel Ticket

IMPLEMENTATION SCREENSHOTS

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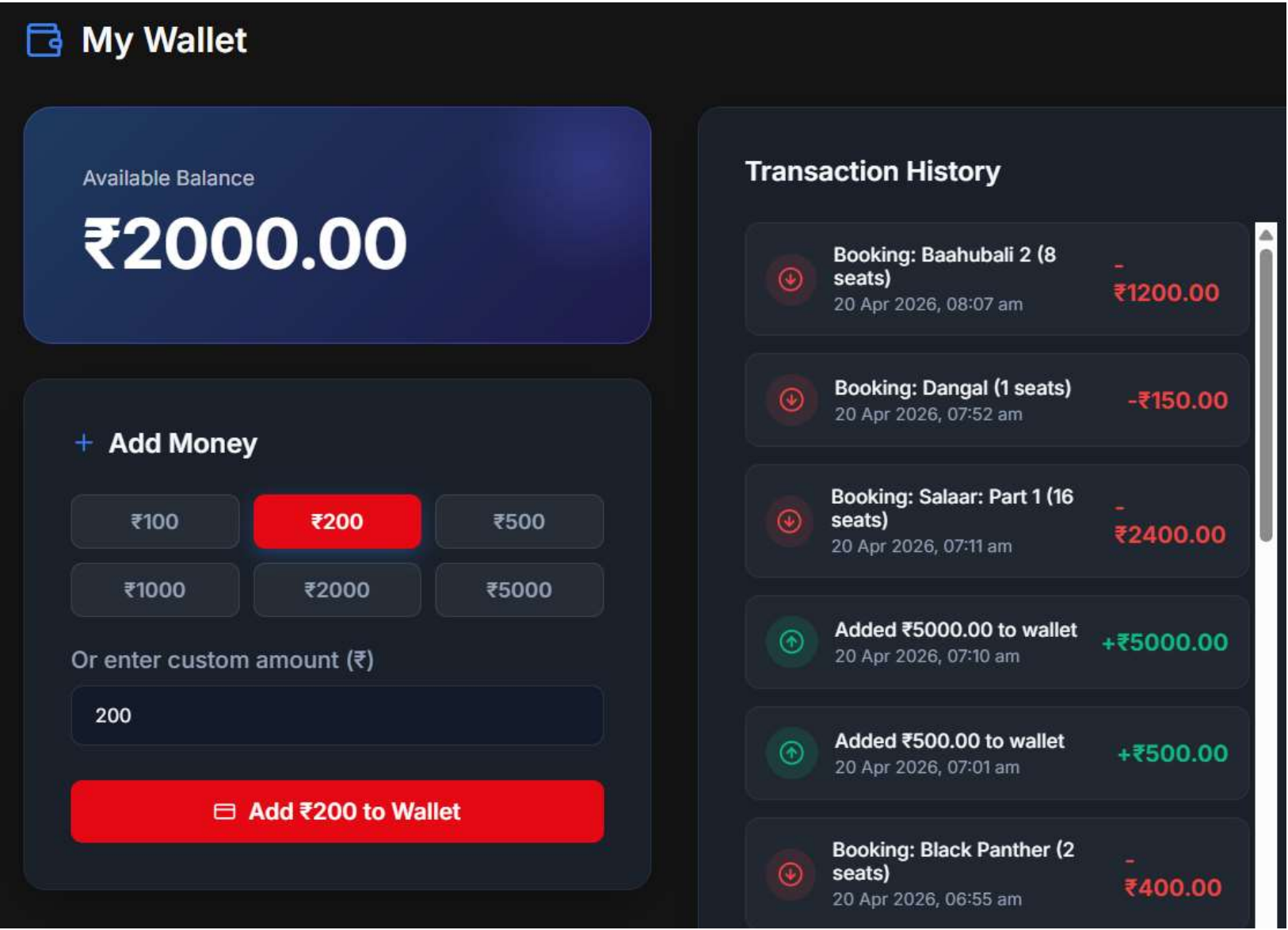
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ADVANTAGES & LIMITATIONS



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Advantages

- Real-time seat availability updates after every booking and cancellation
- JWT-based authentication ensures secure and role-protected access for users and admins
- Modular design makes the system easy to maintain and extend.
- Centralized PostgreSQL database ensures data consistency.

Limitations

- Basic frontend with limited user experience.
- No integrated payment gateway (mock/simulated payments only).
- No real-time seat locking during simultaneous bookings by multiple users.
- Limited analytics and reporting features.

Thank you

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